Name: _____

WORKSHEET

Proof by induction

We know the following pattern is true for numbers we can check.

$$1 = 1^2$$
$$1 + 3 = 2^2$$
$$1 + 3 + 5 = 3^2$$
$$\vdots$$
$$\therefore\ 1 + 3 + 5 + \ldots + (2n - 1) = n^2$$

By programming this pattern in *Excel* we can validate the formula.

	A	B	C	D	E	F	G
1	1	=A1*2 − 1	=B1	=A1^2			
2	2		=C1 + B2				
3	3						
4	4		etc.				
5	5						
6	6						
7	7						
.	.						
.	.						
.	.						

Fill down in columns B, C and D. Columns C and D should replicate the LHS and RHS of the pattern above, as seen on the next page.

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1	1	1	1
2	3	4	4
3	5	9	9
4	7	16	16
5	9	25	25
6	11	36	36
7	13	49	49
8	15	64	64
9	17	81	81
10	19	100	100
11	21	121	121
12	23	144	144
13	25	169	169
14	27	196	196
15	29	225	225
16	31	256	256
17	33	289	289
18	35	324	324
19	37	361	361
20	39	400	400

Excel has its limitations. For what largest value of n can *Excel* validate the formula? Note that the largest number that you can enter or produce in a spreadsheet is dependent upon the version.
This process does not constitute a proof.

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